# Collision system and energy dependence of di-hadron correlations

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# Discussion points

- Track merging – is it clear?  Questions?
- 2D fits update/v3
- Manuel's comments & responses
- Paul – raw correlations avail. on website – format?
- University of Birmingham authors
  - Contributed to track merging correction
  - Leon Galliard, Marek Bombara, Lee Barnby, Peter Jones, John Nelson
  - Request to add them as authors on the paper
- Previous discussion points – next page

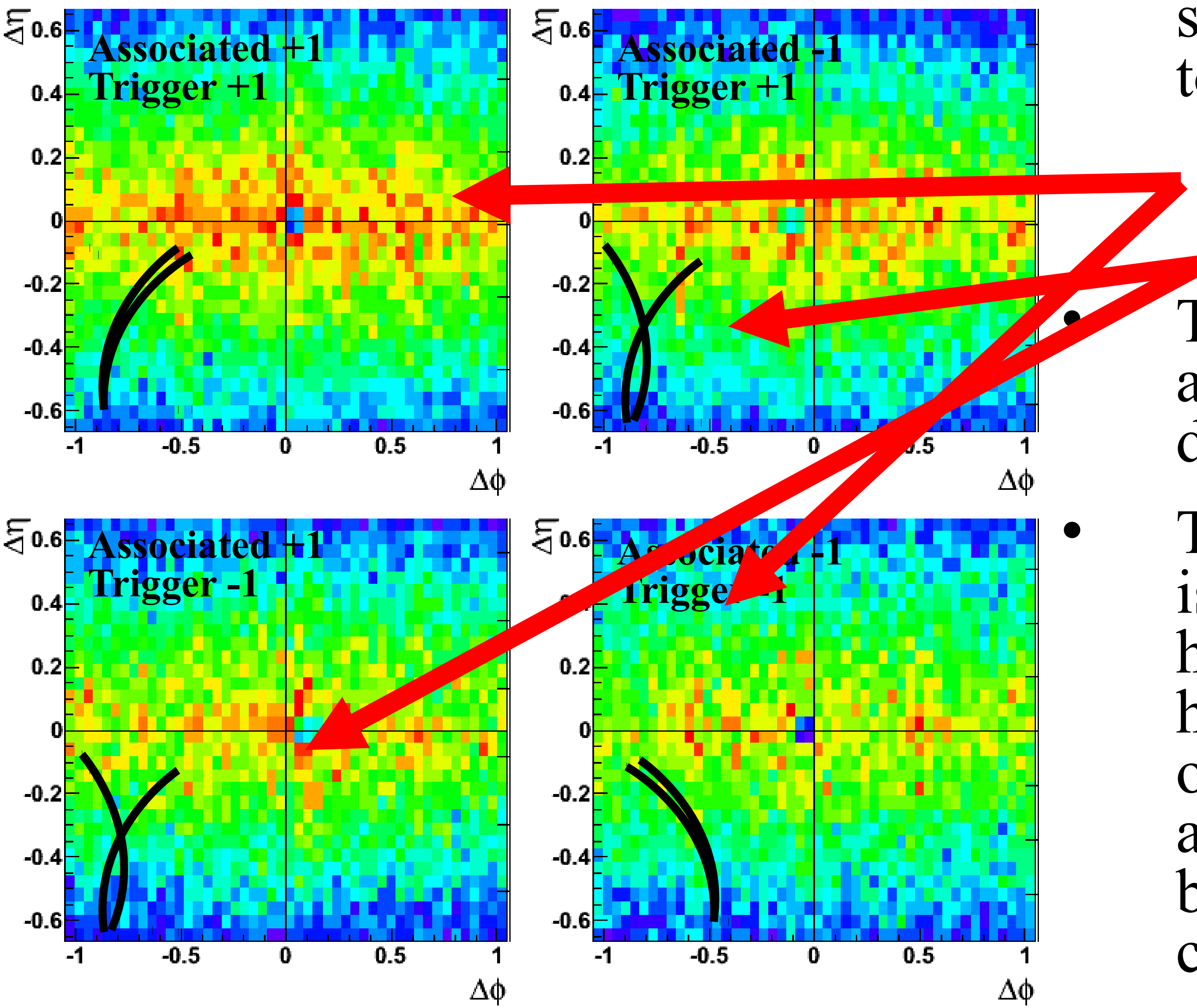
# Previous: Areas open to improvement

- Should the more peripheral Au+Au 62 data be shown?
  - $v_2$ bins are wide, probably leading to overestimate of $v_2$
  - As a result, the error bar includes negative numbers. This is not physical.
- Cu+Cu systematic errors reasonable? - *update, discussion*
- Extend discussion of limitations of ZYAM?
- Need $N_{bin}$ for Au+Au@62 GeV - *update*
- Extend discussion of the models?
- Mention CMS, ISR p+p results in discussion?

Items in blue will not be included in the next discussion unless anyone has something to say.
If no one speaks up we'll assume you're happy with what we've done

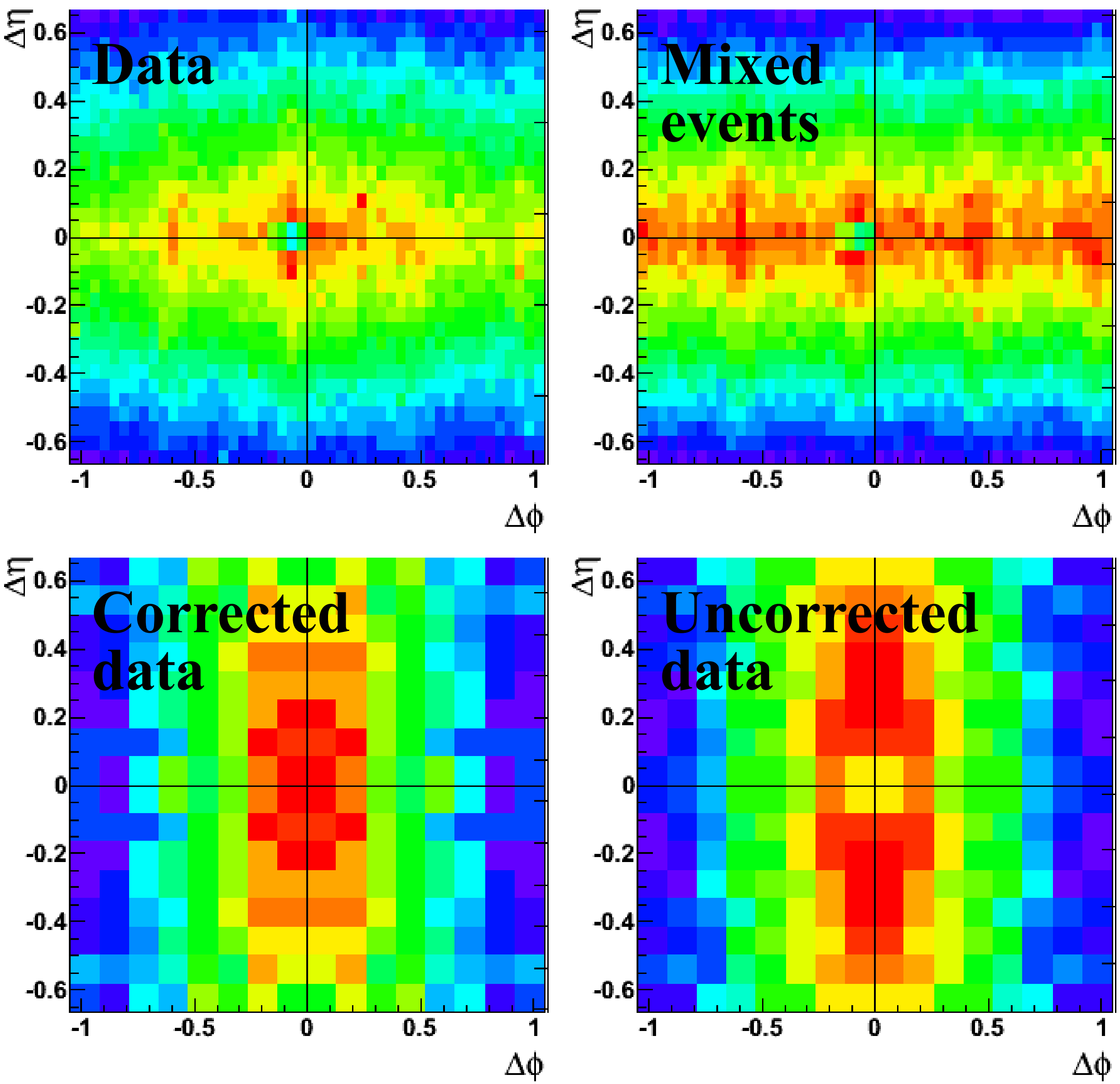
# Track Merging Correction

# Track merging correction




- Track pairs close in space may be lost due to two effects:
  - Track merging
  - Track crossing
- This creates an artificial dip in the distribution of tracks
- The location of this dip is dependent on the helicity
  h = |qB|/qB
  of the trigger and associated particles because of the track curvature

# Track merging correction

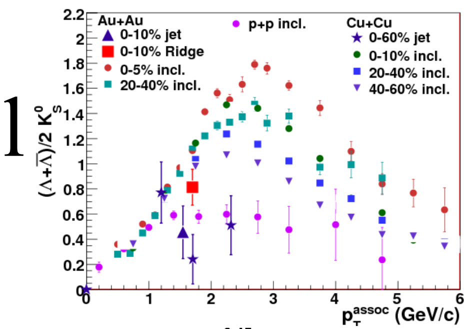


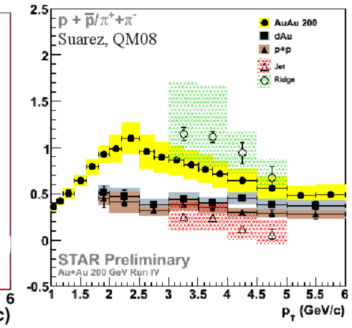

- To correct for lost track pairs:
- Rotate all data so the dip is in the same place
- Recreate the dip in mixed events by calculating the number of merged hits and throwing out all track pairs with >10% merged hits
- Divide the data by the mixed events
- Only do this for small $\Delta\varphi$, $\Delta\eta$ (CPU intensive)
- Patch into full histogram
- Dip in different location for $K^0_S$, $\Lambda$, $\Xi$, $\Omega$. Corrected for each species separately

# Theories

# Key experimental results

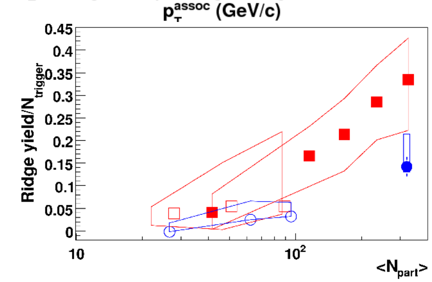
(1) Jet-like correlation is dominantly produced by fragmentation → *Ridge* production must not affect formation of jet-like correlation

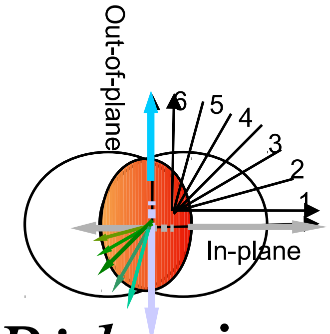



(2) Particle ratios in *Ridge* comparabl
to bulk

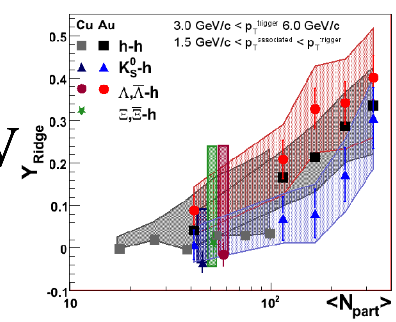
(3) The *Ridge* is smaller in collisions at $\sqrt{s_{NN}}$ = 62 GeV than 200 GeV



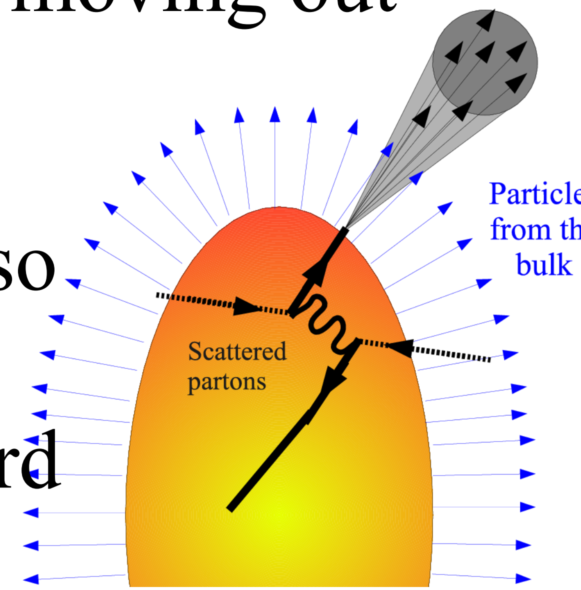

*(4) Ridge* is larger in plane than out of plane

(5) If there is a mass ordering, *Ridge* increases w
increasing trigger mass




(6) The Ridge is broad in Δη

# Radial flow + trigger bias




- Radial flow means that most particles are moving out from the surface of the medium
- If hard partons are also surface biased, hadrons from fragmenting partons will also be surface biased
- This will lead to a correlation between hard partons and medium partons → the *Ridge*
- Several implementations of this mechanism
    - A boost in momentum mimicking radial flow added to PYTHIA (Voloshin, Pruneau, Gavin)
    - Radial flow + trigger bias added to a Glasma initial state (Gavin, Moschelli, McLerrin)
    - Correlated Emission Model adds this mechanism to Recombination (Hwa)

# Radial flow + trigger bias

(1) Jet-like correlation: not affected ☺

(2) Particle ratios: from bulk → comparable to bulk ☺

(3) Ridge in 62 vs 200 GeV: not clear 😐

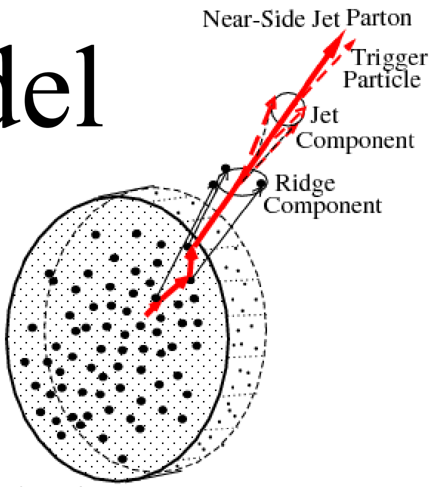
(4) Reaction plane dependence: described by data ☺

(5) Mass ordering: If surface bias is from radial flow, ridge should be larger for lighter trigger particles because lighter particles have greater flow. The opposite trend is observed → if this is the mechanism, surface bias must come from jet quenching 😐

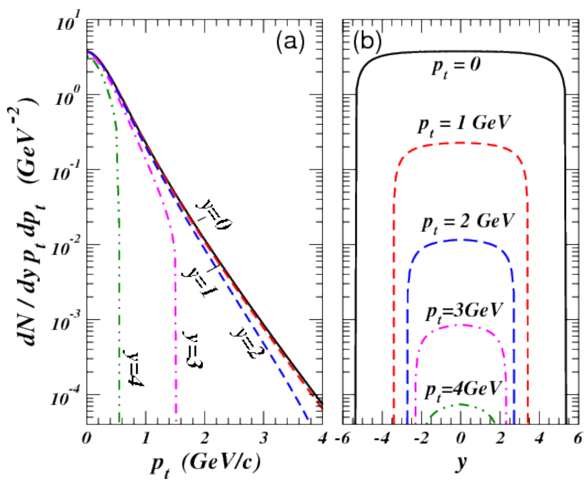
(6) Ridge is broad in $\Delta\eta$: described by model ☺

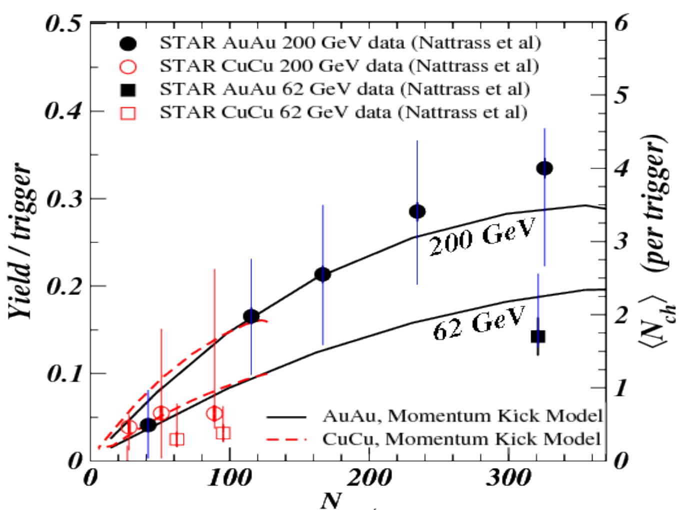
→ Consistent with most observations <u>but</u> need more quantitative calculations

# Momentum kick model




- Hard partons moving through the medium collide with partons from the medium
- These partons acquire a momentum kick in the direction of the hard parton, leading them to be correlated with the hard parton → the *Ridge*
- Since the distribution of medium partons is broad in Δη, the *Ridge* is broad in Δη
- Causal limit to how far in Δη the correlation in Δη can extend → unusual distribution in Δη, strongly $p_T$ dependent
- Able to describe energy dependence



# Momentum kick model

(1) Jet-like correlation: not affected ☺

(2) Particle ratios: from bulk → comparable to bulk ☺

(3) *Ridge* in 62 vs 200 GeV: agrees with data ☺

(4) Reaction plane dependence: naively greater for longer path length ☹

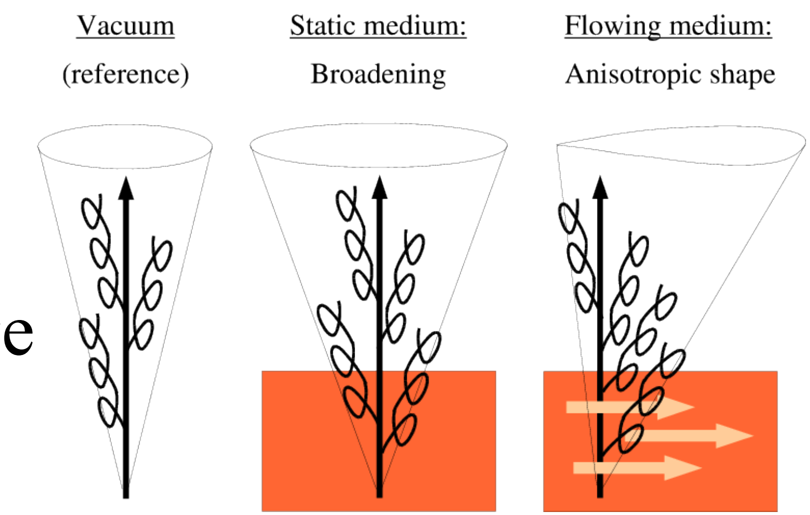
(5) Mass ordering: unclear 😐

(6) Ridge is broad in $\Delta\eta$: described by model ☺

→ Consistent with most observations <u>but</u> needs to be reconciled with reaction plane dependence

# Gluon radiation

- Hard partons moving through the medium emit gluon bremsstrahlung
- Longitudinal flow:
  - Longitudinal flow sweeps these gluons in pseudorapidity →the Ridge




- Plasma instabilities
  - Plasma instabilities form due to the rapidly fluctuating strong fields early in the formation of the collision
  - These fluctuating strong fields make the plasma unstable and deflect gluons radiated by hard partons

# Gluon radiation

(1) Jet-like correlation: not affected ☺

(2) Particle ratios: from gluon fragmentation, likely comparable to ratios in p+p ☹

(3) *Ridge* in 62 vs 200 GeV: not clear 😐

(4) Reaction plane dependence: naively greater for longer path length ☹

(5) Mass ordering: For longitudinal flow mechanism, also expect greater *Ridge* for lighter particles because flow is greater. ☹

(6) *Ridge* is broad in $\Delta\eta$: not consistent with current calculations, may have causal problems ☹

→ Several aspects not likely consistent with data